# Laptop bag vs. Purse: Design a classifier

- Features:
  - Width
  - Height
  - Weight
  - Embeddings

- Classifier: threshold/MLP/loss function

# Evaluation Metrics

Let the problem statement be *classifying purse and bags*.
Purses are labeled as positive class and bags are labeled as negative class

| | | Predicted Class | |
|---|---|---|---|
| | | **Negative** | **Positive** |
| **Actual Class** | **Negative** | A (true negative) | C (false positive) |
| | **Positive** | D (false negative) | B (true positive) |

| Term | Meaning | Example |
|---|---|---|
| True positive | Correct classification | Purse identified as purse |
| False positive | Incorrect classification | Bag identified as purse |
| True negative | Correct classification | Bag identified as bag |
| False negative | Incorrect classification | Purse identified as bag |

http://www.cs.rpi.edu/~leen/misc-publications/SomeStatDefs.html

# Evaluation Metrics

| | | Predicted Class | |
|---|---|---|---|
| | | **Negative** | **Positive** |
| **Actual Class** | **Negative - 50** | A (true negative) - 40 | C (false positive) - 10 |
| | **Positive - 50** | D (false negative) - 20 | B (true positive) - 30 |

| | | **Negative** | **Positive** |
|---|---|---|---|
| **Actual Class** | **Negative - 95** | A (true negative) - 90 | C (false positive) - 5 |
| | **Positive - 5** | D (false negative) - 3 | B (true positive) - 2 |

| Metric | Formula |
|---|---|
| Average classification accuracy | [(TN + TP) / (TN + FP + TP + FN)] |
| Class-wise classification accuracy | [TN / (TN + FP) + TP / (TP + FN)]/2 |
| Type I error (false positive rate) | FP / (TN + FP) |
| Type II error (false negative rate) | FN / (FN + TP) |
| True positive rate | TP / (TP + FN) |
| True negative rate | TN / (TN + FP) |



http://www.cs.rpi.edu/~leen/misc-publications/SomeStatDefs.html

# Evaluation Metrics

| Metric | Formula |
|---|---|
| Average classification accuracy | (TN + TP) / (TN+TP+FN+FP) |
| Type I error (false positive rate) | FP / (TN + FP) |
| Type II error (false negative rate) | FN / (FN + TP) |
| True positive rate | TP / (TP + FN) |
| True negative rate | TN / (TN + FP) |

- Type I error or false positive rate: The chance of incorrectly classifying a (randomly selected) sample as positive
- Type II error or false negative rate: The chance of incorrectly classification a (randomly selected) sample as negative

# Evaluation Metrics

| Metric | Formula |
|---|---|
| Average classification accuracy | (TN + TP) / (TN+TP+FN+FP) |
| Type I error (false positive rate) | FP / (TN + FP) |
| Type II error (false negative rate) | FN / (FN + TP) |
| True positive rate | TP / (TP + FN) |
| True negative rate | TN / (TN + FP) |

- Type I error or false positive rate: The chance of incorrectly classifying a (randomly selected) sample as positiv
- Type II error or false negative rate: The cha classification a (randomly selected) samp

Prevalent in computer vision and image processing related classification problems

# Evaluation Metrics

| Metric | Formula |
|---|---|
| Precision | TP / (TP + FP) |
| Recall | TP / (TP + FN) |

Precision: Fraction of retrieved instances that are relevant

Recall: Fraction of relevant instances that are retrieved

# Evaluation Metrics

| Metric | Formula |
|---|---|
| Precision | TP / (TP + FP) |
| Recall | TP / (TP + FN) |

Precision: Probability that a (randomly selected) retrieved document in relevant

Recall: Probability that a (randomly selected) relevant document is retrieved in a search

# Evaluation Metrics

| Metric | Formula |
|---|---|
| Precision | TP / (TP + FP) |
| Recall | TP / (TP + FN) |

Precision: Probability that a (randomly selected) retrieved document in relevant

Recall: Probability that a (randomly selected) relevant document is retrieved in a search

More prevalent in information retrieval domain

# Evaluation Metrics

| Metric | Formula |
|---|---|
| Sensitivity | TP / (TP + FN) |
| Specificity | TN / (TN + FP) |
| Predictive value for a positive result (PV+) | TP / (TP + FP) |
| Predictive value for a negative result (PV-) | TN / (TN + FN) |

Sensitivity: Proportion of actual positives which are correctly identified

Specificity: Proportion of actual negatives which are correctly identified

Sensitivity: The chance of correctly identifying positive samples. A sensitive test helps rule out disease (when the result is negative)

Specificity: The chance of correctly classifying negative samples. A very specific test rules in disease with a higher degree of confidence.

# Evaluation Metrics

| Metric | Formula |
|---|---|
| Sensitivity | TP / (TP + FN) |
| Specificity | TN / (TN + FP) |
| Predictive value for a positive result (PV+) | TP / (TP + FP) |
| Predictive value for a negative result (PV-) | TN / (TN + FN) |

Sensitivity: The chance of correctly identifying positive samples. A sensitive test helps rule out disease (when the result is negative)

Specificity: The chance of correctly classifying negative samples. A very specific test rules in disease with a higher degree of confidence.



# Evaluation Metrics

| Metric | Formula |
|---|---|
| Sensitivity | TP / (TP + FN) |
| Specificity | TN / (TN + FP) |
| Predictive value for a positive result (PV+) | TP / (TP + FP) |
| Predictive value for a negative result (PV-) | TN / (TN + FN) |

Predictive value of a positive result: If the test is positive, what is the probability that the patient actually has the disease

Predictive value of a negative result: If the test is negative, what is the probability that the patient does not have the disease

# F1 Score

- The F1 score is the harmonic mean of precision and recall:

$$F_1 = \frac{2}{\frac{1}{precision} + \frac{1}{recall}}$$

- Unlike regular mean, harmonic mean gives more weight to low values.
- Therefore, the classifier's F1 score is only high if both recall and precision are high.

# Performance Evaluation

- Classification is of two types:
  - Authentication / verification (1:1 Matching)
    - Is she Richa?
    - Is this an image of a helicopter?
  - Identification (1:n matching)
    - Who's photo is this?
    - This image belongs to which class?

# Practice Question

- The Delhi Traffic Police has deployed an AI-based traffic violation detection system at a busy intersection in Connaught Place to automatically detect vehicles running red lights. The system captures images and classifies each vehicle as either a "Violator" or "Non-Violator" in real time.
  To evaluate the system, 1,000 vehicles passing the intersection over a 3-hour period were manually verified by human inspectors, and their results were compared to the system's predictions.
- The following table summarises the findings:

| Actual Condition | Predicted as Violator | Predicted as Non-Violator |
|---|---|---|
| Violator | 420 | 80 |
| Non-Violator | 60 | 440 |

**Violator** = Vehicle actually ran a red light.
**Non-Violator** = Vehicle obeyed the traffic signal.

# Practice Question

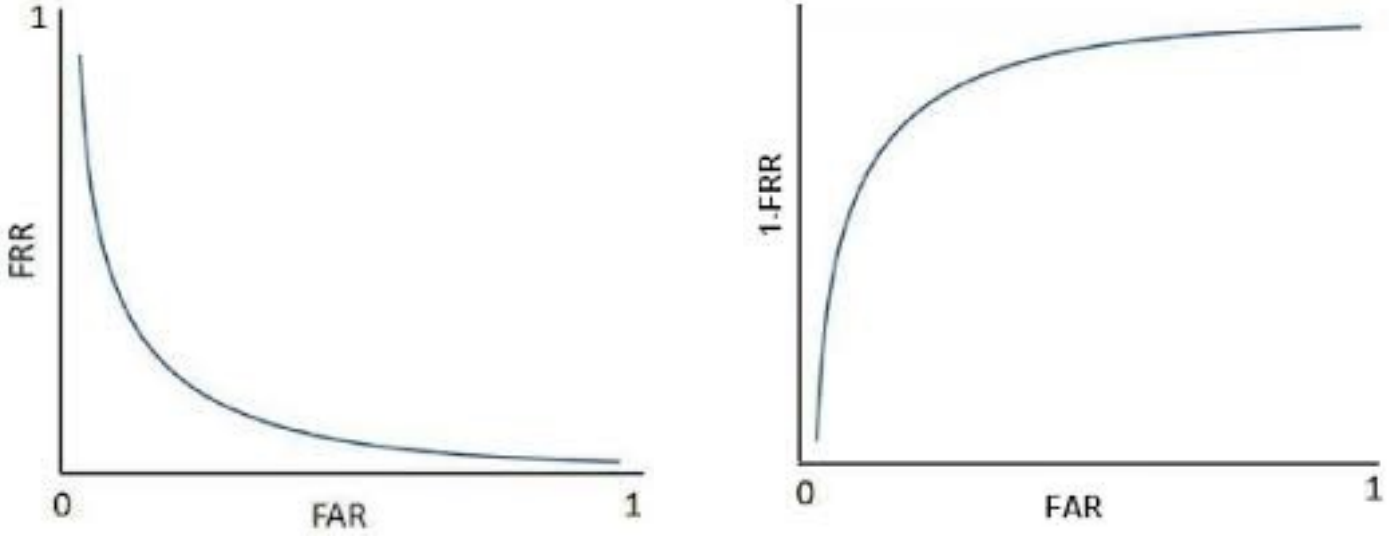
- Compute the TPR, TNR, FPR and FNR

- **TPR** = TP / (TP + FN) = 420 / 500 = **0.84 = 84%**
  Fraction of real violators the system catches.
- **FPR** = FP / (FP + TN) = 60 / 500 = **0.12 = 12%**
  Fraction of law-abiding vehicles wrongly flagged.
- **TNR (Specificity)** = TN / (FP + TN) = 440 / 500 = **0.88 = 88%**
  Fraction of non-violators correctly cleared.
- **FNR** = FN / (TP + FN) = 80 / 500 = **0.16 = 16%**
  Fraction of violators the system misses.

# Performance Evaluation

- Receiver operating characteristics (ROC) curve
  - For authentication/verification
  - False positive rate vs true positive rate
  - False accept rate vs true accept rate
- Detection error-tradeoff (DET) curve
  - False positive rate vs false negative rate
  - False accept rate vs false reject rate
- Cumulative match curve (CMC)
  - Rank vs identification accuracy

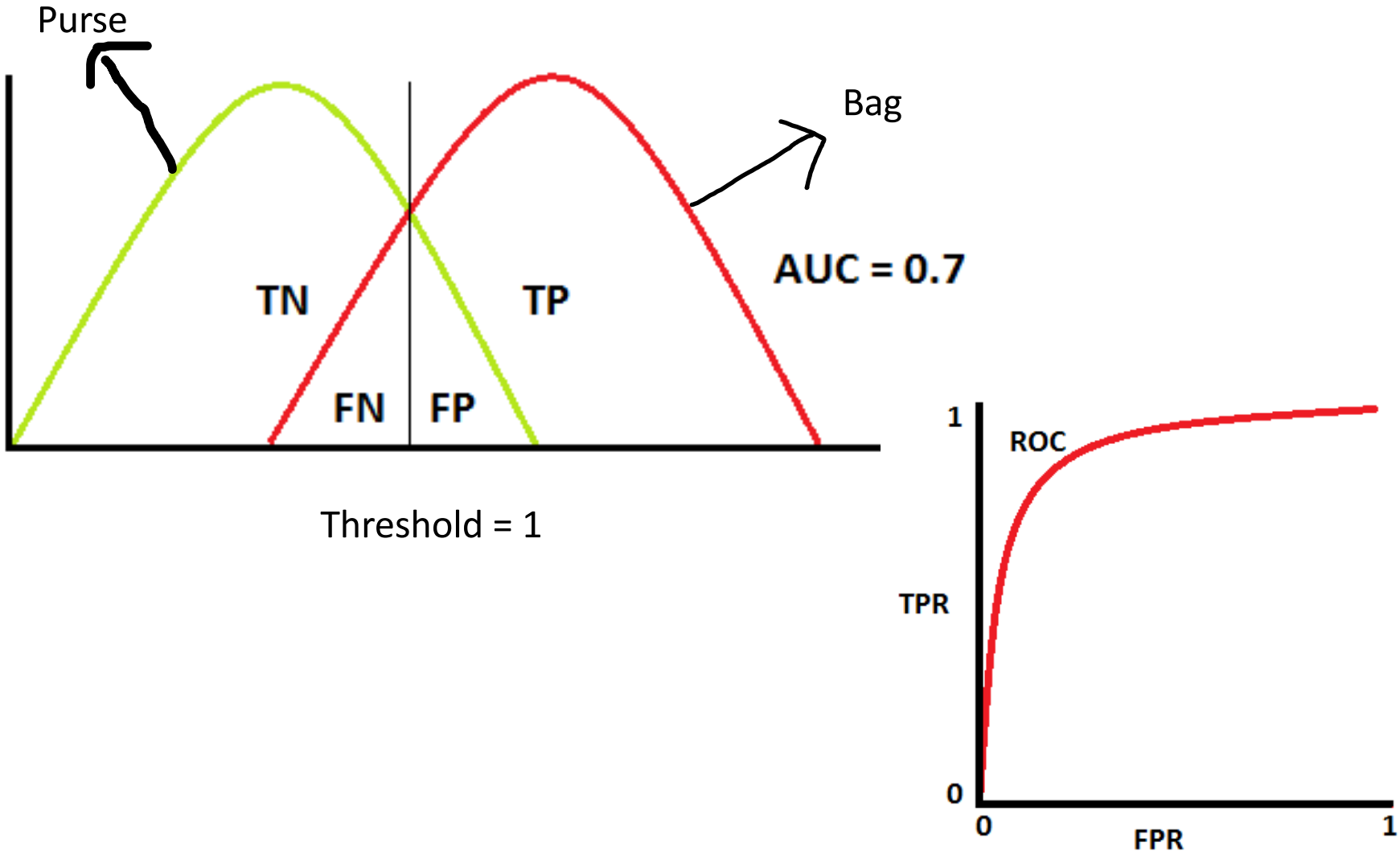
# ROC Curve



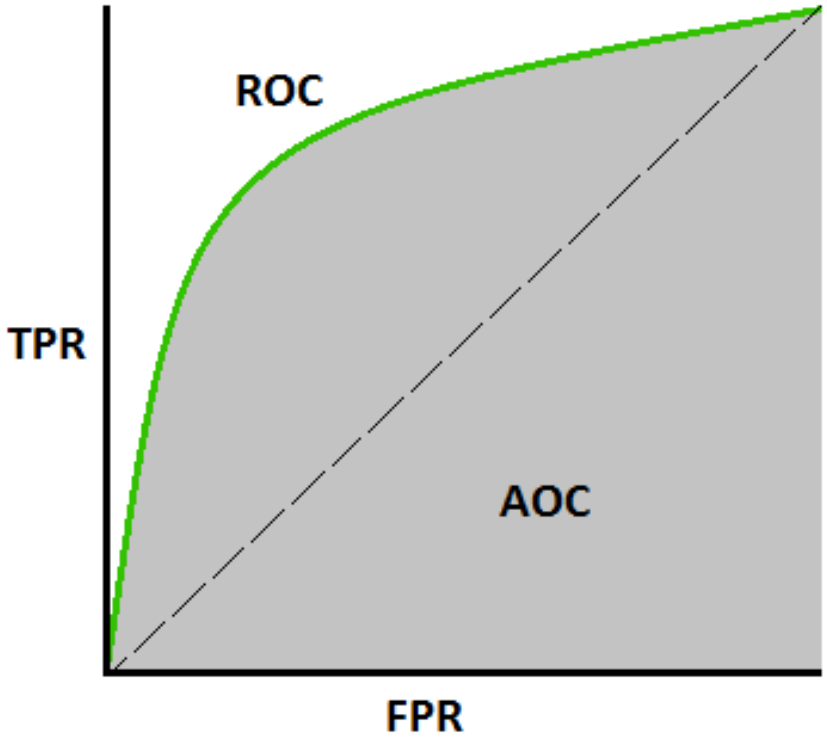
# Laptop bag vs. Purse: Design a classifier

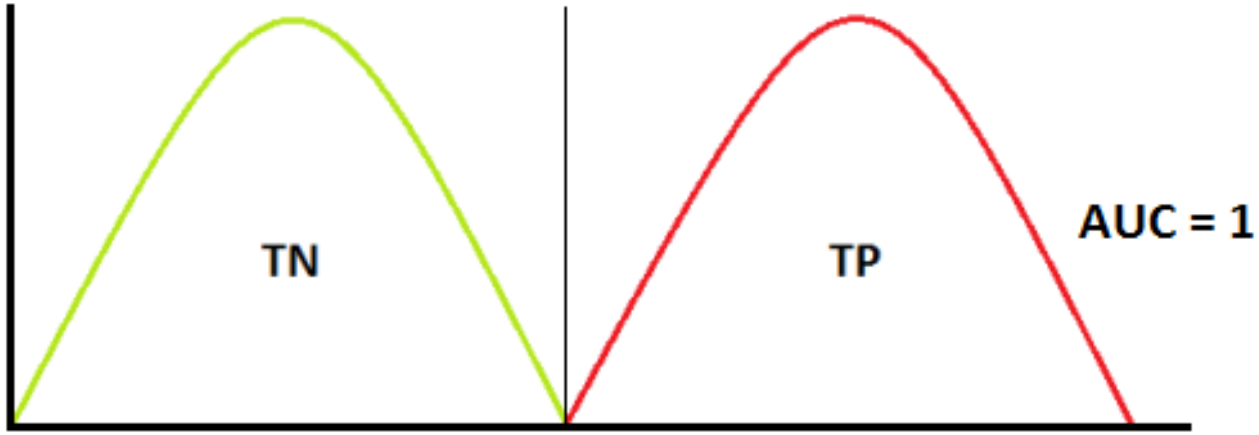
- Features:
  - Ratio of Height to Width

- Classifier: threshold

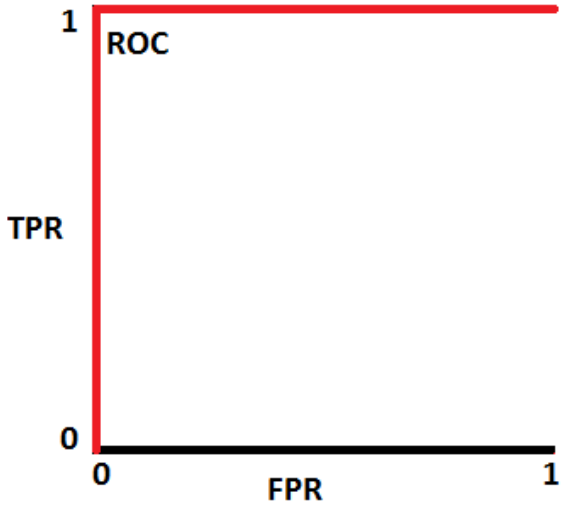

Purse
Bag
AUC = 0.7
TN
TP
FN
FP
Threshold = 1
1
ROC
TPR
0
0
FPR
1

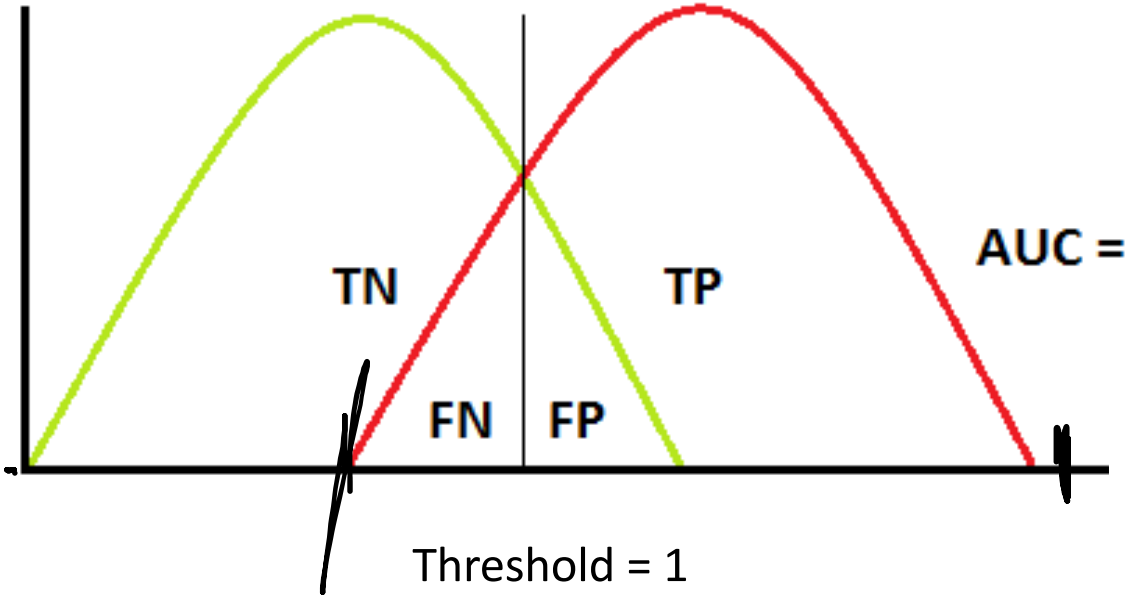
# Area Under the Curve

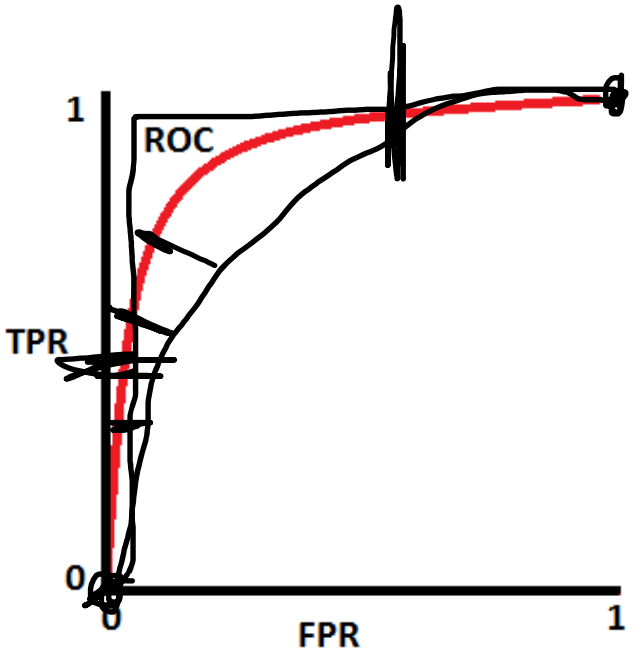


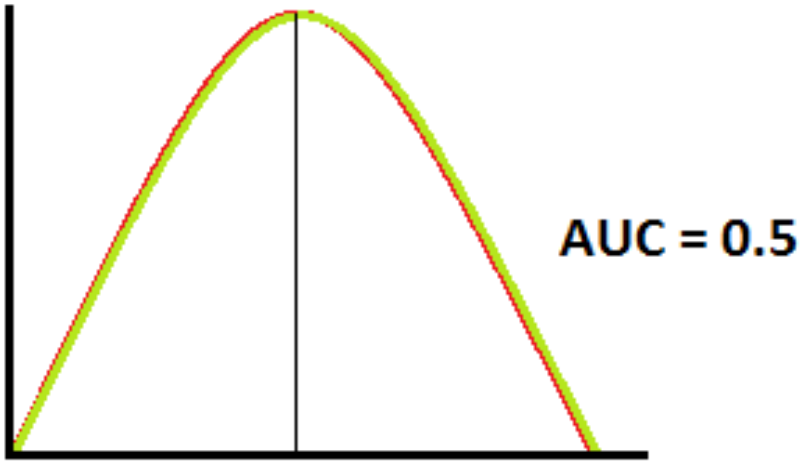



Threshold = 1

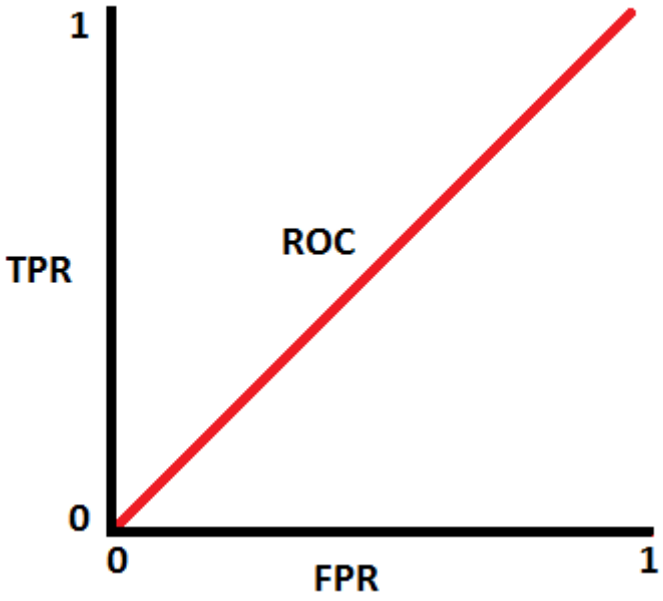


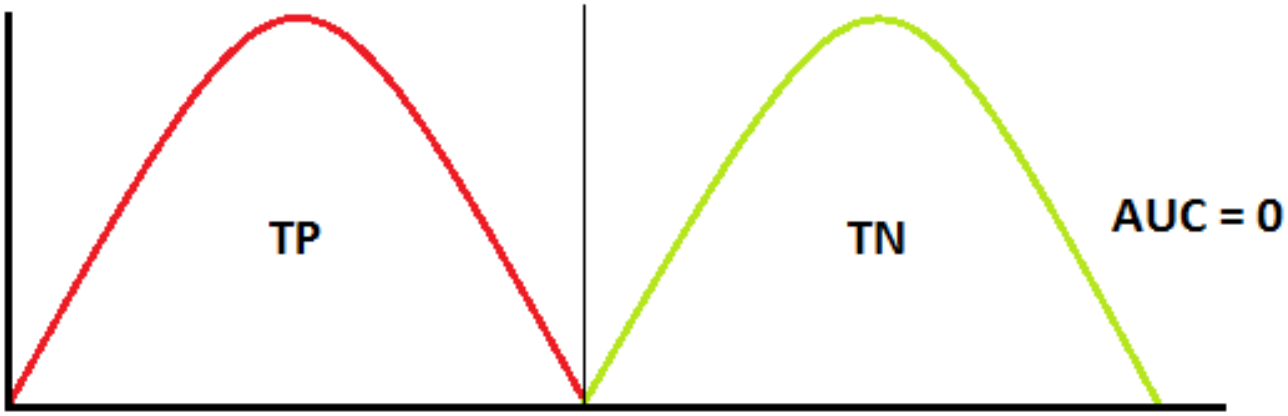

TN
TP
FN
FP
AUC = 0.7
Threshold = 1


1
ROC
TPR
0
0
FPR
1

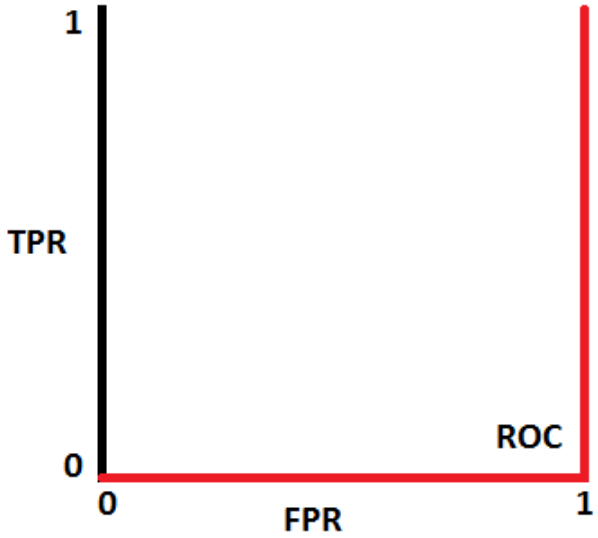



Threshold = 1






Threshold = 1



# Sensitivity vs. Specificity

